

4.11.4 Data Summarization and Statistical Analysis

گروپ بائی ایک جیسے ریکارڈ جوڑتا ہے، پھر COUNT, SUM, AVG, MAX, MIN سے خلاصہ نکلتا ہے۔

CS XI

Chapter 4: Data and Analysis

Lecture 14 of 15

4.11.4 Data Summarization and Statistical Analysis

LEARN IT

Grouping combines rows that share a value (all CS students together) and then an **aggregate** calculates a number for the group. In Access click Totals (Σ) to reveal the Total row in the grid.

Function	What it calculates
COUNT()	How many records (students in IT)
SUM()	Total of a number field (fees collected)
AVG()	Mean
MAX() / MIN()	Highest / lowest in the group

WORKED EXAMPLE

Average marks per department: add Department and Marks; Totals Σ ; Department $\rightarrow$ Group By; Marks $\rightarrow$ Avg; Run.

Department	Avg of Marks
Computer Science	82.5
IT	76.0

Check: CS marks 80, 85 $\rightarrow$ average $(80+85)/2 = 82.5$. IT 70, 82 $\rightarrow 76.0$.

```
SELECT Department, AVG(Marks) AS AvgOfMarks
FROM Student
GROUP BY Department;
```

اردو میں سمجھیں

گروپ بائی ایک جیسے ریکارڈ جوڑتا ہے، پھر COUNT, SUM, AVG, MAX, MIN سے خلاصہ نکلتا ہے۔ ایکسیس میں Σ بٹن Total قطار کھولتا ہے۔

CHECK YOURSELF

1. Which Total setting goes under Department?
2. Which function counts how many books each member has borrowed?

Answers: 1. Group By. 2. COUNT (of BorrowID or BookID) grouped by MemberID.

Next lecture: 4.11.6 Data Visualization in MS Access (Charts)